

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27
Name	A. Priyanka	
Roll Number	24011101001	

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image
Reconstructed image
Reconstruction error

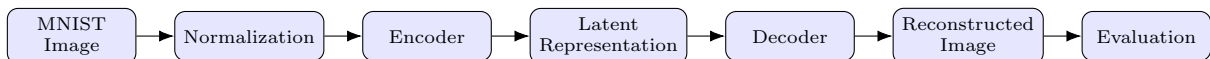
For the denoising model, display:

Original clean image
Noisy image
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

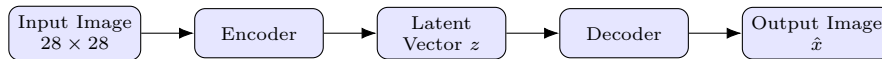
$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images

The trained fully connected autoencoder was evaluated on unseen test images. The figure shows the original images, their reconstructed outputs, and the corresponding absolute reconstruction error maps.

FC-AE: Original | Reconstruction | Absolute Error Map

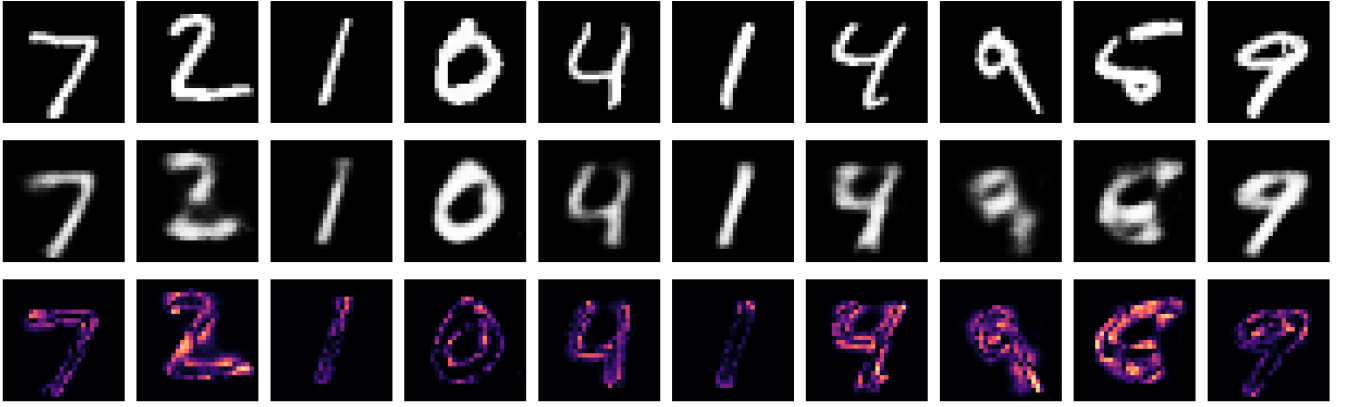


Figure 1: Plot 1: Original images, FC-AE reconstructions, and absolute reconstruction error maps.

Inference: The reconstructed digits preserve the overall shape and identity of most samples. However, some fine details and thin strokes are slightly blurred, which is expected from the compressed latent representation of the fully connected autoencoder. The error maps show that reconstruction errors are mainly concentrated around digit boundaries and regions with fine or irregular strokes. Simpler and thicker digits are reconstructed more clearly, whereas thin or irregular digits show comparatively larger errors.

Plot 2: Training and Validation Reconstruction Loss

The training and validation reconstruction losses were recorded over the 20 training epochs.

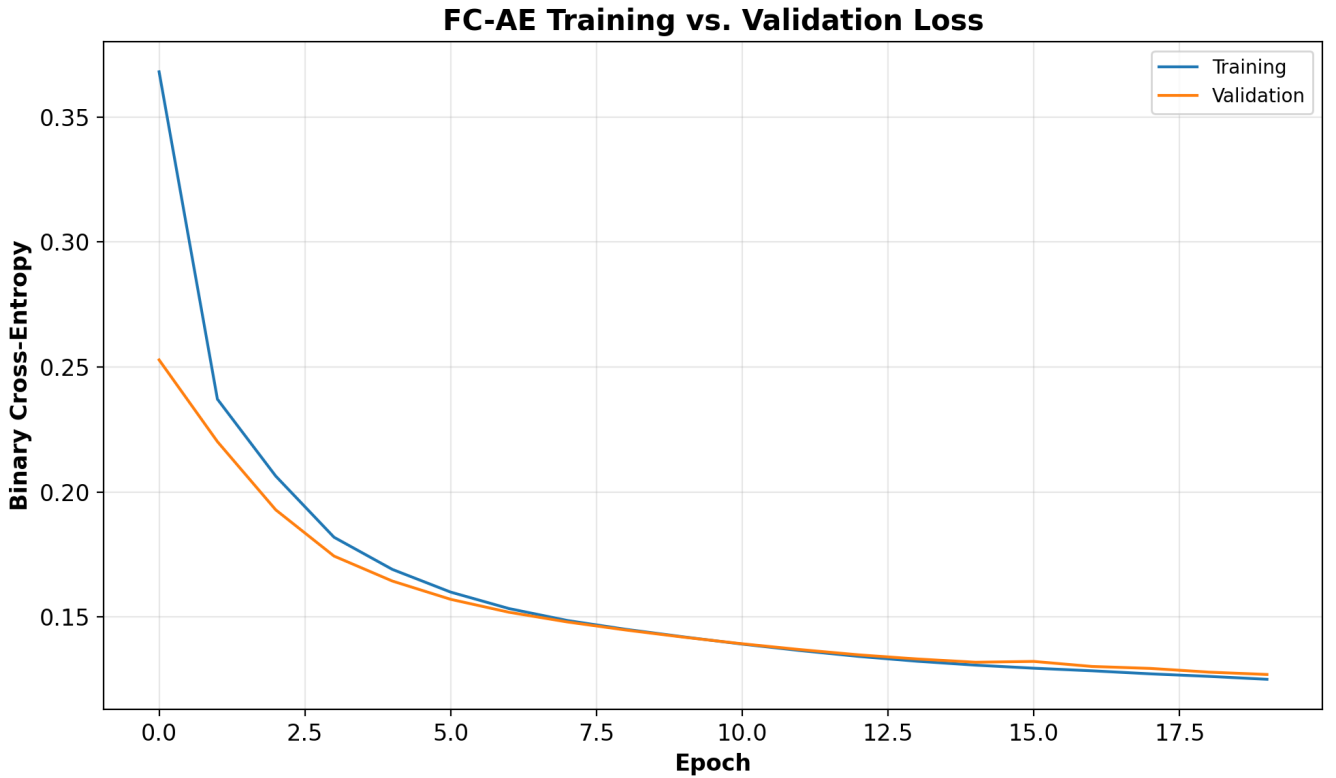


Figure 2: Plot 2: Training and validation reconstruction loss of the FC-AE.

Inference: Both training and validation losses decrease rapidly during the initial epochs and gradually flatten towards the end of training. This indicates that the autoencoder is learning to reconstruct the input images and is approaching convergence. The training and validation curves remain close throughout the later epochs, with no significant divergence, indicating no strong evidence of overfitting. The model therefore shows stable convergence over the 20 epochs.

9. Reconstruction Metrics

The final FC-AE was evaluated on the untouched test set using Mean Squared Error (MSE), Mean Absolute Error (MAE) and Structural Similarity (SSIM).

Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2.$$

Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|.$$

Structural Similarity

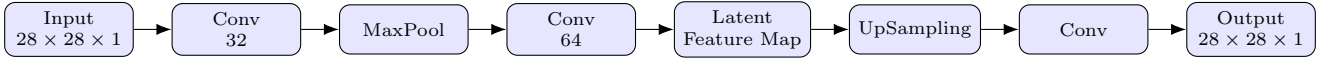
SSIM measures the structural similarity between the original and reconstructed images.

Metric	Value
Test MSE	0.02239
Test MAE	0.05884
Mean SSIM	0.7328

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```
Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

11. Comparing Fully Connected and Convolutional Autoencoders

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.02239	0.05884	0.7328	211040
Convolutional AE	0.00259	0.01511	0.9731	74497

Plot 3: Reconstruction Comparison

Plot 3: Original | FC-AE | CAE

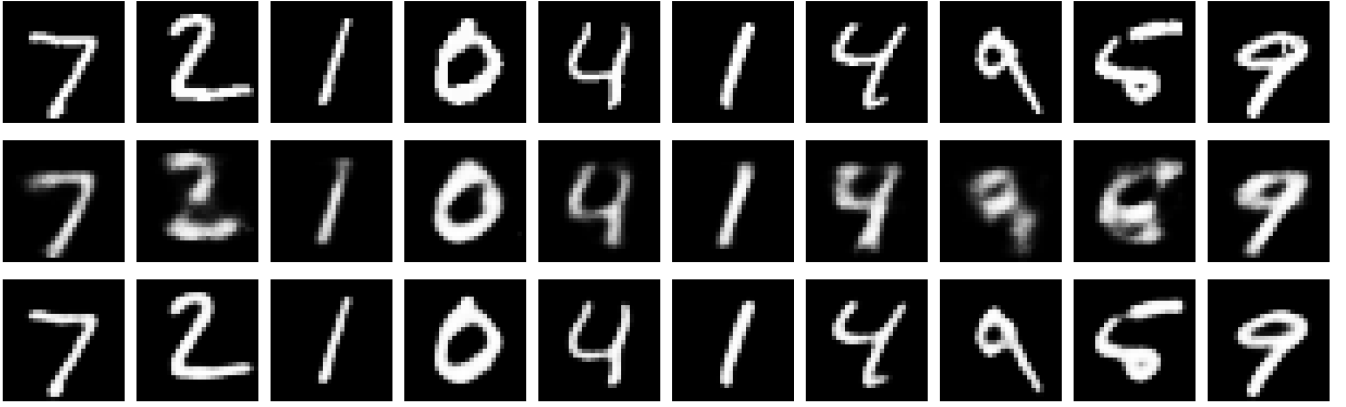


Figure 3: Plot 3: Original, FC-AE and convolutional-AE reconstructions.

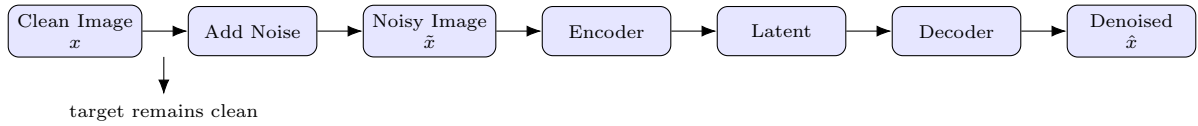
Inference: The convolutional-AE reconstructions are noticeably sharper and closer to the original digits than the FC-AE reconstructions. The CAE preserves digit edges and local details better which shows the advantage of convolution in capturing spatial patterns.

12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image. The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where $\tilde{x}$ is a noisy version of the clean image x .



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.

Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

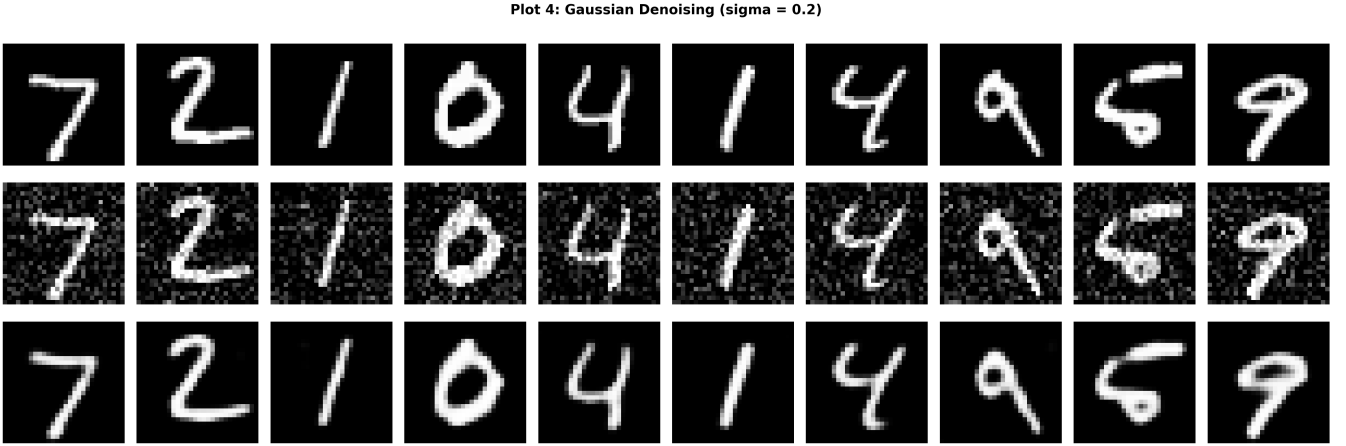


Figure 4: Plot 4: Clean, Gaussian-noisy and denoised test images ($\sigma = 0.2$).

Inference: The denoising autoencoder removes most of the added Gaussian noise while preserving the main shapes of the digits. The denoised images are much cleaner than the noisy inputs although a small amount of noise remains.

Plot 5: Gaussian Noise Level vs. Reconstruction Quality

Noise Level σ	MSE	MAE	SSIM
0.1	0.00372	0.01793	0.95836
0.2	0.00464	0.02128	0.94468
0.3	0.00710	0.02971	0.86519

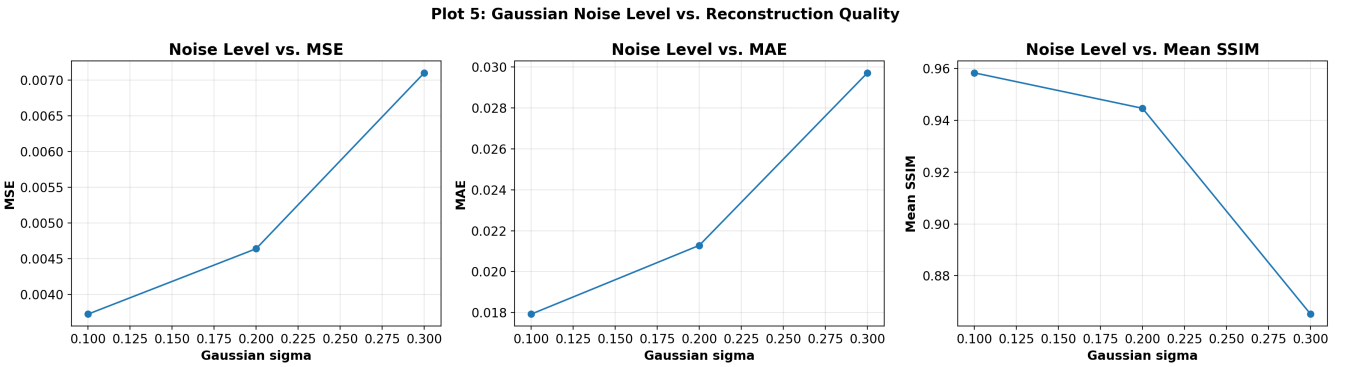


Figure 5: Plot 5: Gaussian noise level versus MSE, MAE and SSIM.

Inference: As the noise level increases, MSE and MAE increase while SSIM decreases. This shows that stronger Gaussian noise makes it harder for the denoising autoencoder to recover the original image accurately.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

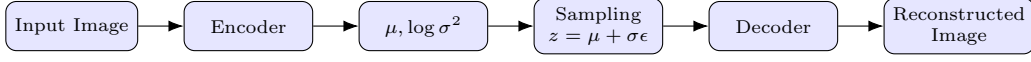
$$q_{\phi}(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produces:

$$\mu, \quad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}(q_\phi(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The total loss is

$$\boxed{\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

Plot 9: VAE Training and Validation Reconstruction Loss

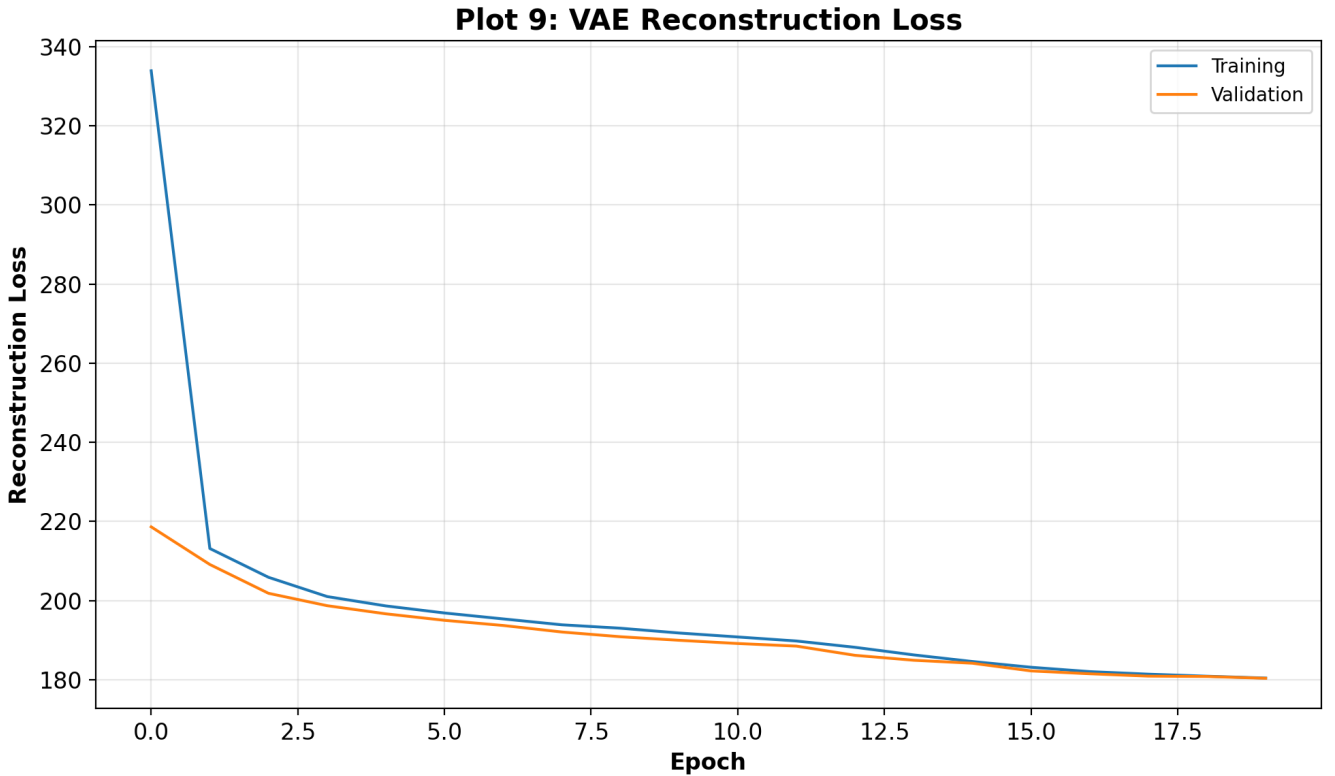


Figure 6: Plot 9: Training and validation reconstruction loss of the VAE.

Inference:

The training and validation reconstruction losses decrease as the VAE learns to reconstruct the input images. The decrease occurs because the encoder and decoder weights are progressively optimized to reduce reconstruction error. The curves become more stable towards the later epochs, showing that the model is approaching convergence. The close behaviour of the two curves indicates no strong evidence of overfitting.

17. VAE Implementation

Use a small latent space so that it can be visualized.

Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

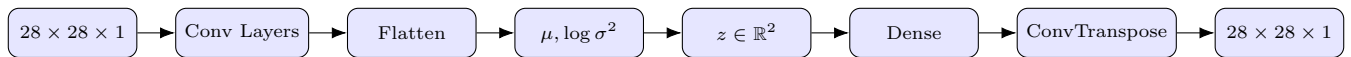
The encoder should produce:

$$\mu \in \mathbb{R}^2, \quad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \rightarrow 28 \times 28 \times 1.$$

A compact architecture is:



18. Exploring the VAE Latent Space

Plot 6: VAE Latent Space

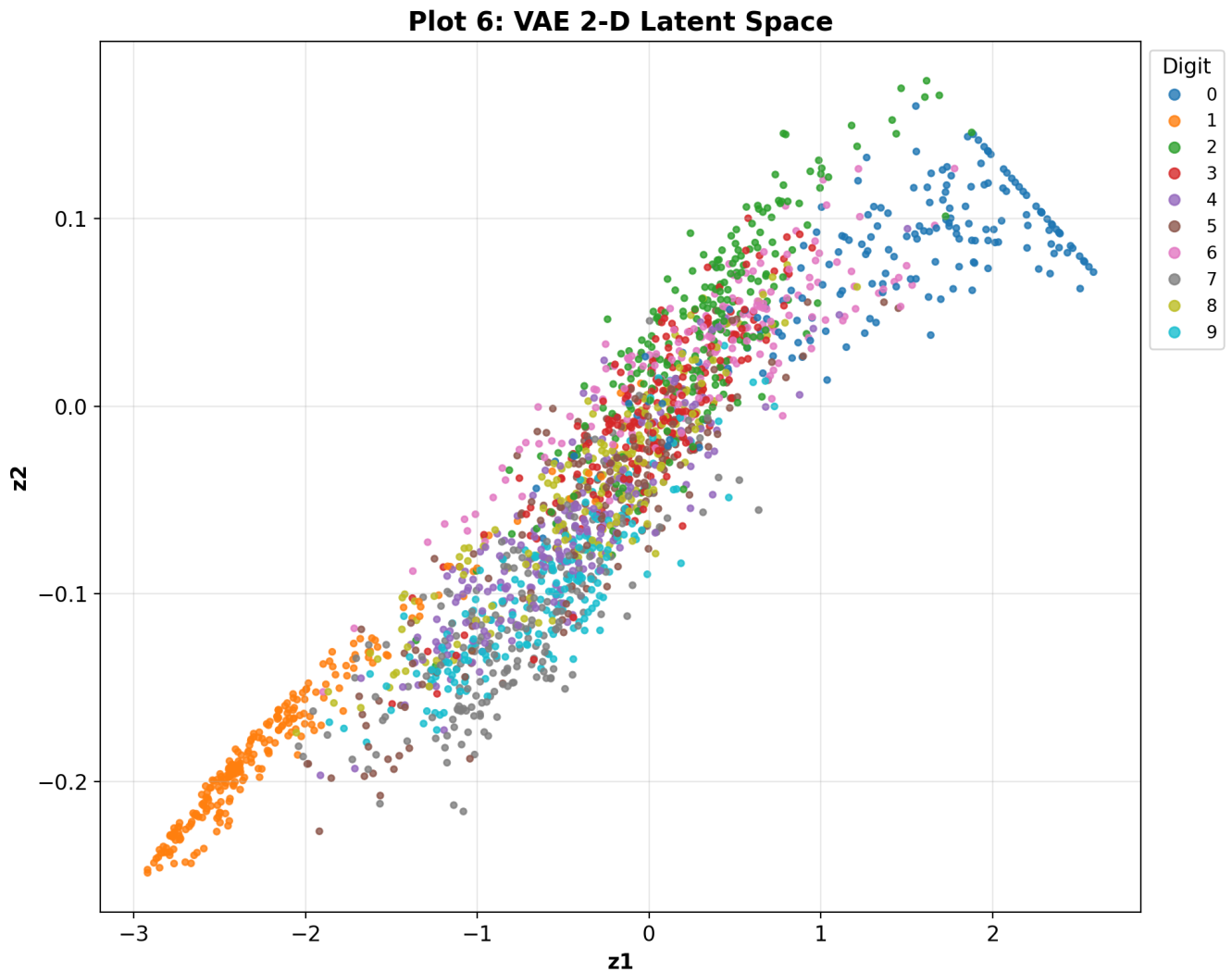


Figure 7: Plot 6: Two-dimensional VAE latent space coloured by digit label.

Inference: The latent points form some visible clusters, but several digit classes still overlap. This shows that the VAE has learned a meaningful latent structure, although a 2D latent space cannot completely separate all digit classes.

19. Generating New Images with the VAE

Plot 7: Randomly Generated Images

Plot 7: 25 Randomly Generated VAE Images

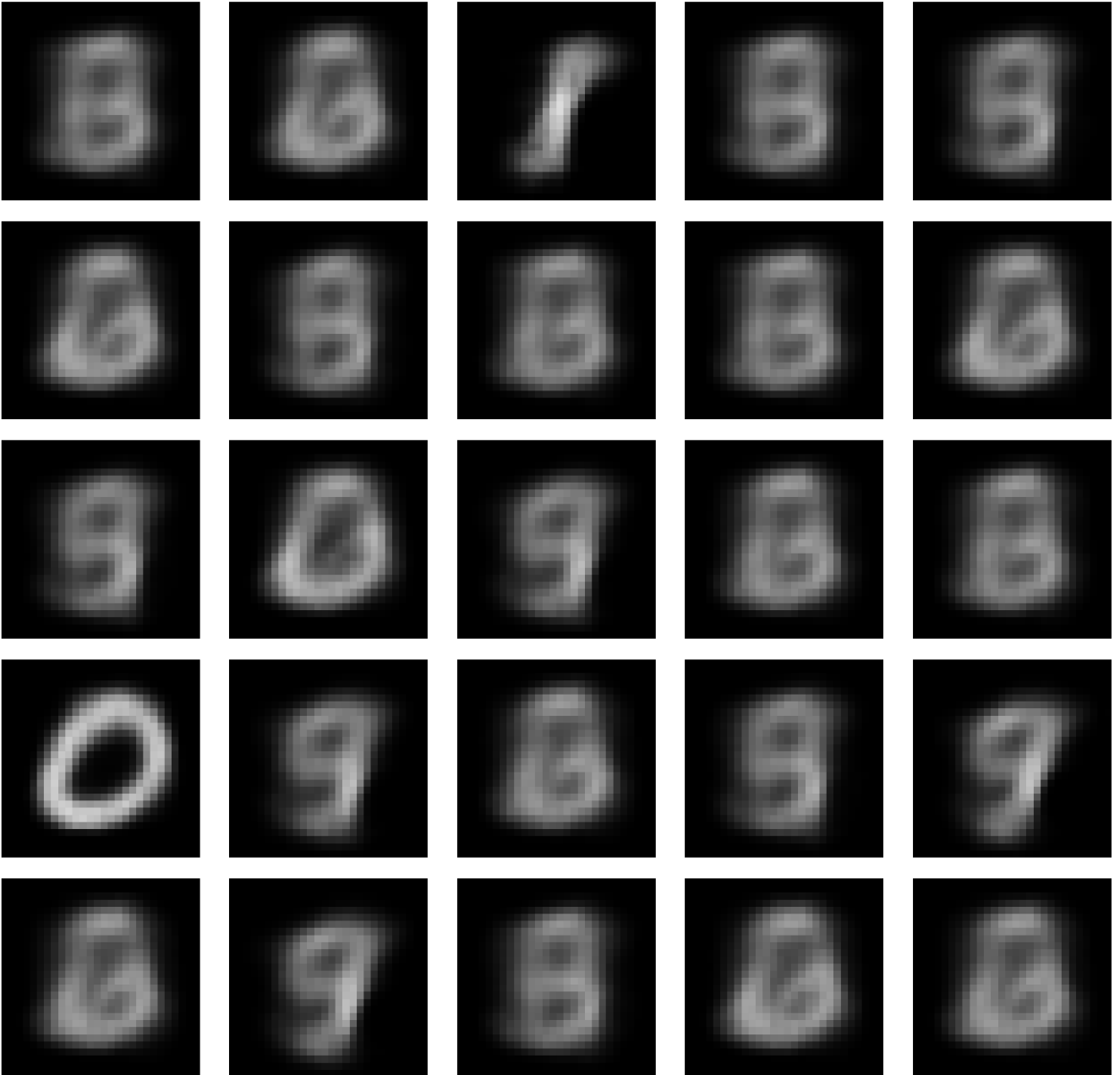


Figure 8: Plot 7: 25 images generated by sampling from the VAE latent prior.

Inference: The generated samples show noticeable variation and generally resemble handwritten digits. Some samples are blurry or ambiguous, which is expected because the main VAE uses a compact two-dimensional latent representation. Overall, the decoder is able to generate new and reasonably diverse digit-like images from randomly sampled latent vectors.

20. Latent Space Interpolation

Plot 8: Latent Space Interpolation

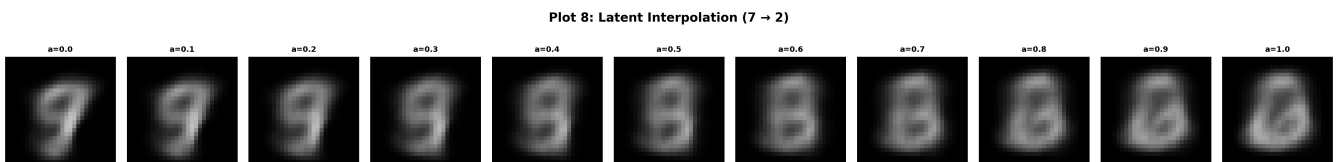


Figure 9: Plot 8: Decoded images obtained by interpolating between two test-image latent vectors.

Inference: The decoded images change gradually as the interpolation parameter moves from 0 to 1, with the digit shape evolving smoothly between the two endpoints. This shows that the VAE has learned a continuous latent space where intermediate latent vectors can produce meaningful intermediate images.

21. Reconstruction Comparison Across Models

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.02239	0.05884	0.7328	211040
Convolutional AE	0.00259	0.01511	0.9731	74497
Denoising CAE	0.00468	0.02138	0.9442	74497

For the VAE, reconstruction and KL terms are reported separately because its objective contains both components.

VAE Metric	Value
Reconstruction Loss	177.3696
KL Loss	2.3794
Total Loss	179.7489
Test MSE	0.05533
Test MAE	0.12939
Mean SSIM	0.3091

22. Reconstruction Error Distribution

The per-image reconstruction error was calculated using the convolutional autoencoder on the test set.

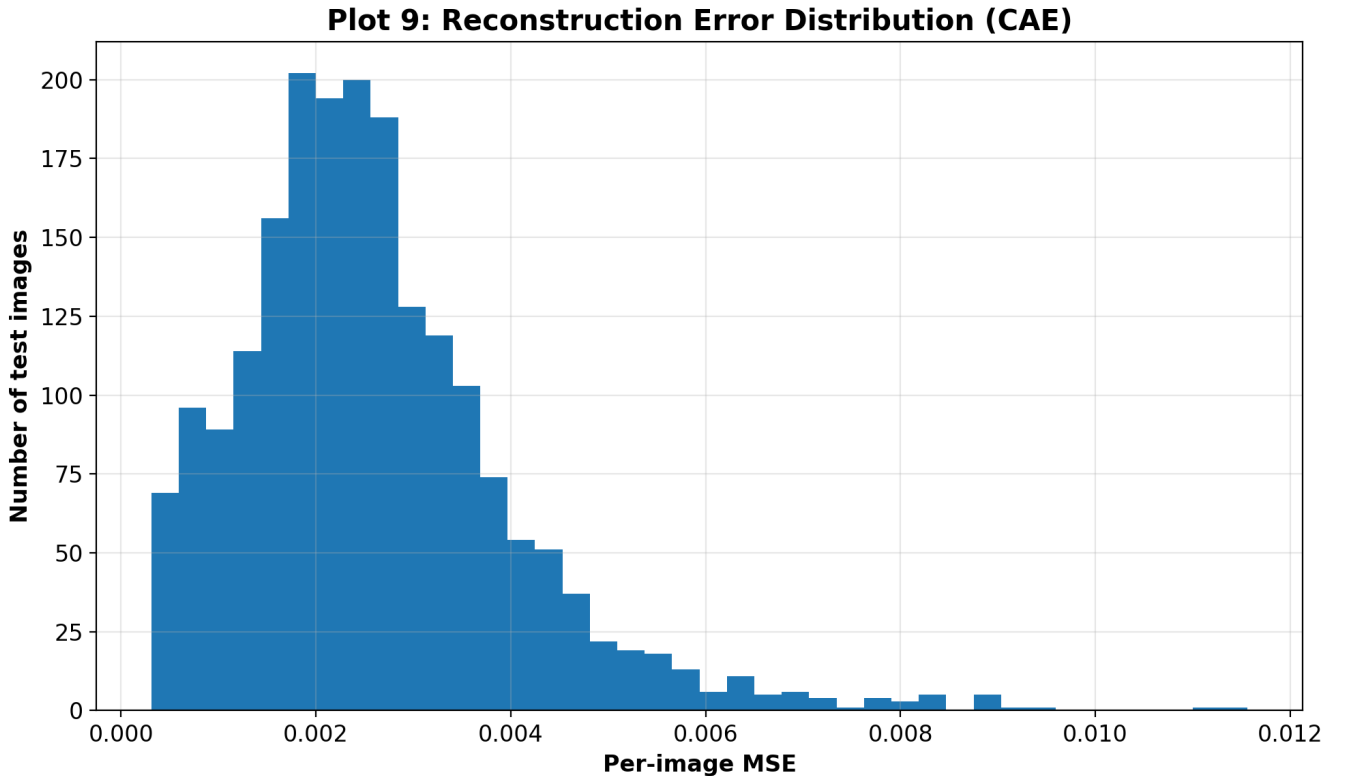


Figure 10: Plot 10: Distribution of per-image reconstruction errors for the CAE.

Inference: Most reconstruction errors are concentrated at the lower end of the distribution, with a smaller number of samples forming a long right tail. This indicates that most test images are reconstructed reasonably well, while a few images with more complex or unclear strokes are harder for the CAE to reconstruct accurately.

23. Required High-Error Image Analysis

The five test images with the largest reconstruction errors were selected automatically from the test set.

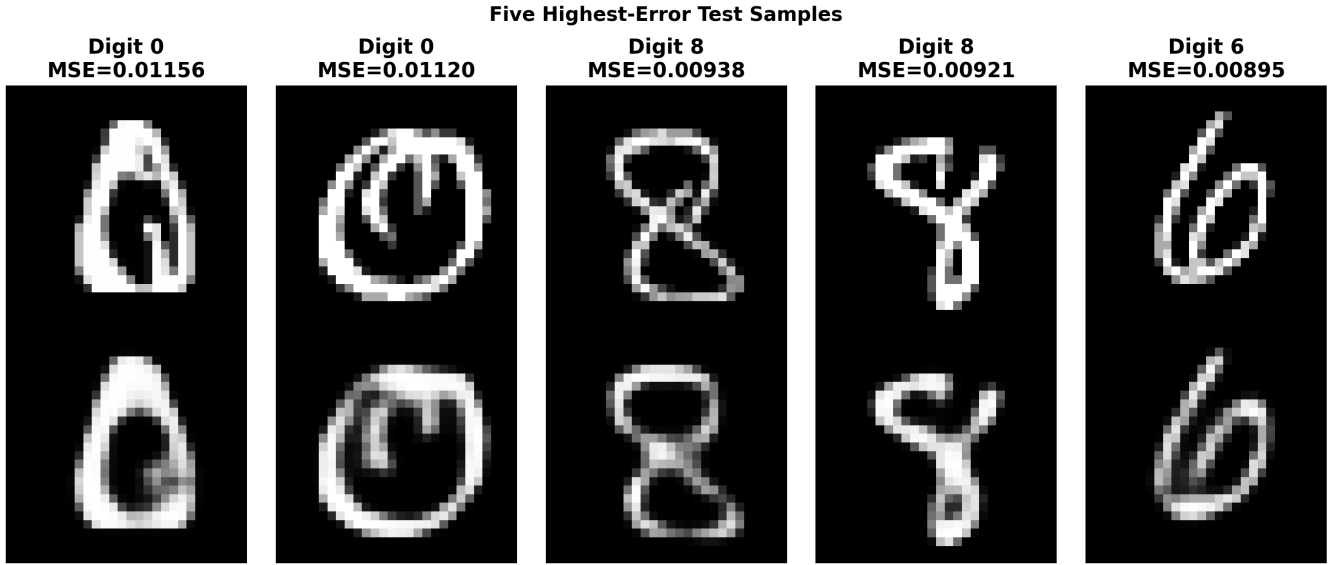


Figure 11: Five test images with the largest CAE reconstruction errors and their reconstructions.

Inference: These samples have noticeably higher reconstruction errors than the majority of the test set. Their irregular shapes and thin or curved strokes make them more difficult for the CAE to reconstruct accurately. The reconstructed images are therefore less clear, which is consistent with their larger reconstruction errors.

24. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time
FC Autoencoder	0.02239	0.05884	0.7328	211040	10.8 s
Conv. Autoencoder	0.00259	0.01511	0.9731	74497	18.4 s
Denoising CAE	0.00468	0.02138	0.9442	74497	15.6 s
VAE	0.05533	0.12939	0.3091	134165	21.0 s

The values were obtained from the final student execution. The denoising CAE row corresponds to Gaussian noise with $\sigma = 0.2$.

25. Required Inferences

- **Effect of latent dimension:**

Increasing the latent dimension allows the autoencoder to retain more information from the input image. The latent-dimension study shows that reconstruction improves as the latent dimension increases, but this comes with a reduction in compression.

- **Fully connected vs. convolutional reconstruction:**

The convolutional autoencoder produces sharper reconstructions than the fully connected autoencoder. By preserving local spatial information, the convolutional model is able to reconstruct digit strokes and edges more accurately.

- **Effect of Gaussian noise level:**

As the Gaussian noise level increases, MSE and MAE increase while SSIM decreases. This shows that stronger noise makes it harder for the denoising autoencoder to recover the original image accurately.

- **Denoising ability:**

The denoising autoencoder removes most of the added noise while retaining the main structure of the digits. However, as the noise becomes stronger, some fine details and thin strokes become harder to recover.

- **VAE latent distribution:**

Unlike a conventional autoencoder, the VAE represents each input using a distribution in the latent space. This gives the model a structured latent representation from which different latent vectors can be sampled and decoded into new images.

- **Structure of the two-dimensional VAE latent space:**

The two-dimensional latent space shows some clustering of digit classes, but several classes still overlap. This indicates that the VAE has learned meaningful patterns in the latent space, although two dimensions are not enough to clearly separate all digit classes.

- **Quality and diversity of generated samples:**

Random samples from the VAE latent space produce different digit-like images with noticeable variation. Some samples are blurry or ambiguous, showing that the VAE can generate new images but that the compact two-dimensional latent space limits the level of detail.

- **Smoothness of latent-space interpolation:**

The decoded images change gradually as the latent vector moves from one endpoint to the other. This shows that intermediate points in the latent space can produce meaningful intermediate image structures.

- **Reconstruction error and visual quality:**

Images with lower reconstruction errors generally have clearer and more accurate reconstructions. The high-error samples show more visible distortions, particularly around unusual shapes, curves, and thin strokes.

- **Deterministic vs. probabilistic latent representations:**

A conventional autoencoder maps an input to a fixed latent vector, while a VAE represents the latent space using probability distributions. This probabilistic representation allows the VAE to sample new latent vectors, generate new images, and smoothly interpolate between latent representations.

26. Additional Latent-Dimension Study

The basic FC-AE was repeated with latent dimensions $d_z \in \{2, 8, 16, 32\}$.

Latent Dimension	MSE	SSIM	Parameters
2	0.04876	0.42116	210130
8	0.02921	0.65543	210520
16	0.02413	0.71282	211040
32	0.02124	0.74179	212080

Latent Dimension vs. Reconstruction MSE

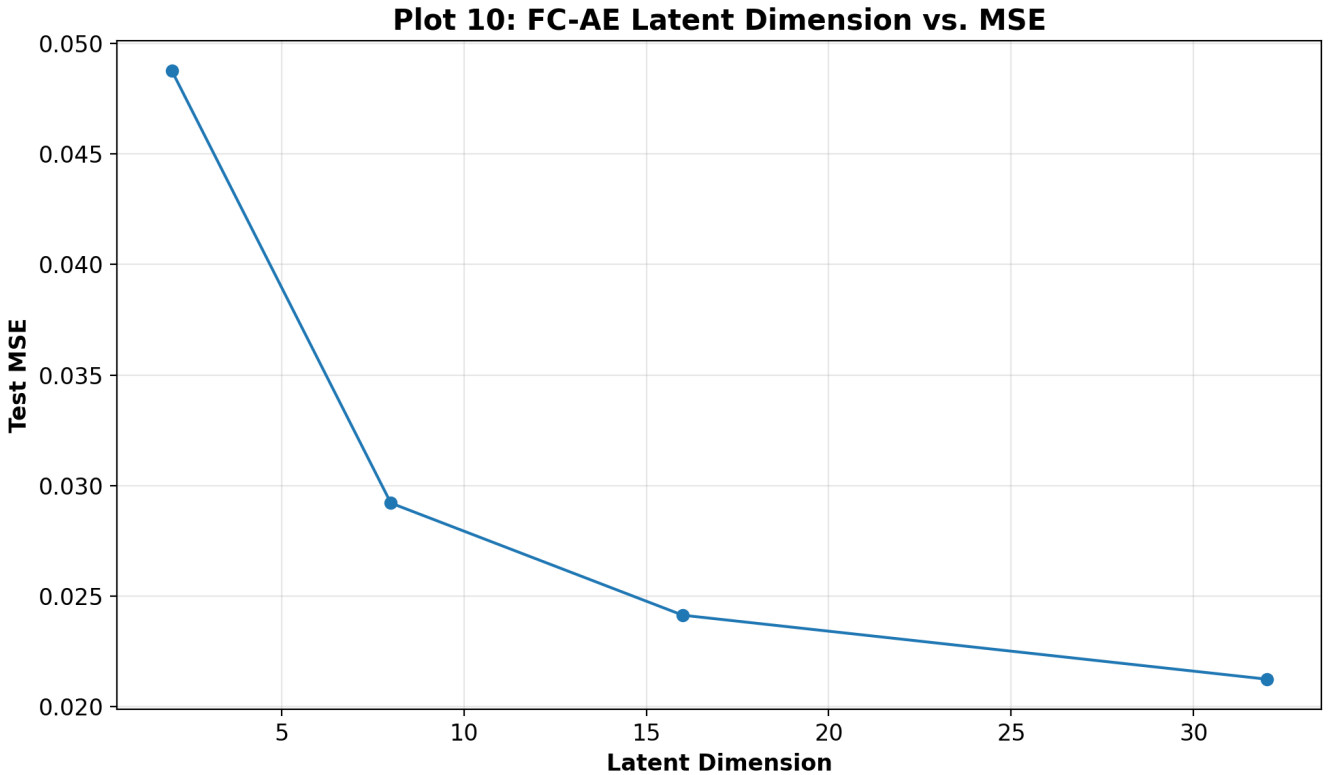


Figure 12: FC-AE latent dimension versus reconstruction MSE.

Inference: Increasing the latent dimension from 2 to 32 reduces the reconstruction MSE and improves SSIM, showing that a larger latent space preserves more image information. However, the improvement becomes smaller

at higher dimensions, while the model provides less compression.

27. Discussion Questions

1. What is an autoencoder?

An autoencoder is a neural network that learns to reproduce its input at the output. It first compresses the input into a smaller representation and then tries to reconstruct the original input from that representation.

2. What is the purpose of the encoder?

The encoder takes the original input and converts it into a smaller latent representation. In simple terms, it tries to keep the important information while reducing the size of the data.

3. What is the purpose of the decoder?

The decoder takes the latent representation produced by the encoder and tries to reconstruct the original input as closely as possible.

4. What is a latent representation?

A latent representation is the compact form of the input produced by the encoder. It contains the main features or information that the network needs to reconstruct the original data.

5. Why is the latent representation usually smaller than the input?

Making the latent representation smaller forces the autoencoder to learn the important features instead of simply copying the input. This is also what gives the autoencoder its compression ability.

6. What happens when the latent dimension is too small?

If the latent dimension is very small, the encoder cannot store enough information about the input. As a result, the decoder has less information available and the reconstructed image usually becomes less accurate.

7. What happens when the latent dimension is increased?

Increasing the latent dimension generally gives the network more space to store useful information. Therefore, reconstruction quality usually improves, although the improvement may become smaller after a certain point.

8. Why can a convolutional autoencoder preserve spatial information more effectively?

Convolutional layers work with local regions of an image and learn spatial patterns such as edges, shapes and textures. Because of this, a convolutional autoencoder can preserve the spatial structure of images better than a fully connected autoencoder.

9. Differentiate an autoencoder and a denoising autoencoder.

A normal autoencoder is trained to reconstruct the original input from itself. A denoising autoencoder is given a noisy version of the input but is trained to reconstruct the original clean image. Therefore, the denoising version learns to remove unwanted noise.

10. Why is the clean image used as the target in denoising?

The clean image is used as the target because the purpose of the model is to learn how to remove the noise. If the noisy image were also used as the target, the network would have no reason to learn the difference between useful image information and noise.

11. What happens when the noise level is increased?

As the noise level increases, more information in the image becomes corrupted. The denoising task therefore becomes harder, and the reconstruction quality generally decreases. In our experiment, higher noise levels resulted in higher reconstruction error and lower SSIM.

12. Why are MSE, MAE and SSIM useful for image reconstruction?

MSE measures the average squared difference between the original and reconstructed pixels, while MAE measures the average absolute difference. SSIM is useful because it also considers structural similarity between the images. Using all three gives a better idea of reconstruction quality than relying on only one metric.

13. What is a Variational Autoencoder?

A Variational Autoencoder (VAE) is an autoencoder that learns a probability distribution in the latent space instead of mapping each input to only one fixed latent vector. This makes it possible to sample new latent points and generate new images.

14. How does a VAE differ from a conventional autoencoder?

A conventional autoencoder produces a fixed latent representation for an input. A VAE learns the mean and variance of a latent distribution and samples from that distribution. Because of this, a VAE can be used for both reconstruction and generation.

15. Why does a VAE learn a distribution instead of a single latent vector?

Learning a distribution makes the latent space more continuous and organised. Nearby points in this space can represent similar data, which makes operations such as sampling and interpolation possible.

16. What is the reparameterization trick?

The reparameterization trick allows the VAE to sample from the learned latent distribution while still allowing backpropagation during training. Instead of directly sampling z , the model uses the mean and standard deviation together with a random value ϵ .

17. Why is ϵ sampled from a standard normal distribution?

ϵ is sampled from a standard normal distribution so that the randomness can be separated from the parameters

learned by the encoder. This gives a simple and consistent way of generating samples from the latent distribution.

18. **What is the role of KL divergence in the VAE loss?**

KL divergence controls how close the learned latent distribution is to the chosen prior distribution. It prevents the latent space from becoming irregular and helps make it suitable for sampling new data.

19. **What is the prior distribution used in the VAE?**

The prior distribution used in this experiment is a standard normal distribution, with mean 0 and variance 1. The KL-divergence term encourages the learned latent distributions to stay close to this prior.

20. **Why is a two-dimensional latent space useful for visualization?**

A two-dimensional latent space can be plotted directly on a graph. This makes it easier to see how different digit classes are distributed and whether similar images tend to appear near each other.

21. **What does latent-space interpolation demonstrate?**

Latent-space interpolation shows that the model can move gradually between two latent representations. When the corresponding decoded images also change gradually, it indicates that the VAE has learned a meaningful and relatively continuous latent space.

22. **Why can a VAE generate new images?**

A VAE learns a structured latent distribution. Once training is complete, a new point can be sampled from the prior distribution and passed through the decoder to produce a new image.

23. **Why might some VAE-generated images be unclear?**

The VAE is learning an approximation of the data distribution rather than memorising individual training images. If the latent representation or decoder is not expressive enough, the generated images can lose fine details and appear blurry or ambiguous.

24. **What is the difference between reconstruction and generation?**

Reconstruction means taking an existing input image, encoding it and then decoding it back into an approximation of the same image. Generation starts with a newly sampled latent point and uses the decoder to create an image that was not directly given as input.

25. **Why should test images remain unseen during training?**

The test set should remain unseen so that it provides an unbiased measure of how well the trained model works on new data. If test images are used during training, the final evaluation may not accurately represent the model's generalisation ability.

28. Additional Exercises

The following additional experiments were performed to study the effect of latent dimensionality, noise type and level, decoder architecture and variational autoencoder regularization.

28.1 Exercise 1: Effect of Latent Dimension in the Convolutional Autoencoder

The convolutional autoencoder was trained with latent dimensions of 4, 8, 16 and 32. The reconstruction quality was compared using MSE, MAE, SSIM and the number of trainable parameters.

Latent Dimension	MSE	MAE	SSIM	Parameters
4	0.03477	0.08358	0.61528	65797
8	0.02097	0.05546	0.76219	90889
16	0.01115	0.03462	0.88056	141073
32	0.00658	0.02488	0.92998	241441

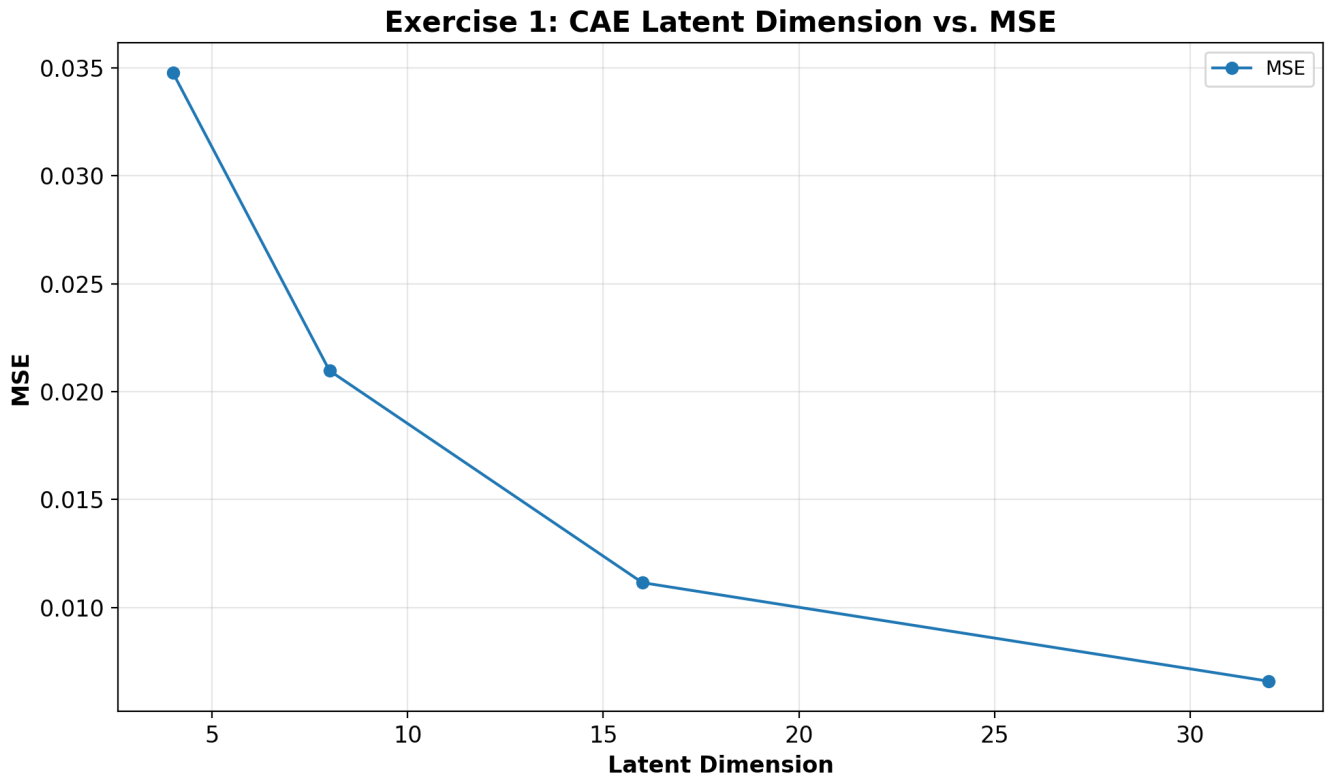


Figure 13: Reconstruction MSE for different latent dimensions of the convolutional autoencoder.

Inference: Increasing the latent dimension from 4 to 32 steadily improves the reconstruction quality, with MSE decreasing from 0.03477 to 0.00658. The largest improvement is seen at the lower latent dimensions, while the gains become smaller as the dimension increases. This shows that a larger latent space helps the CAE preserve more information and produce better reconstructions.

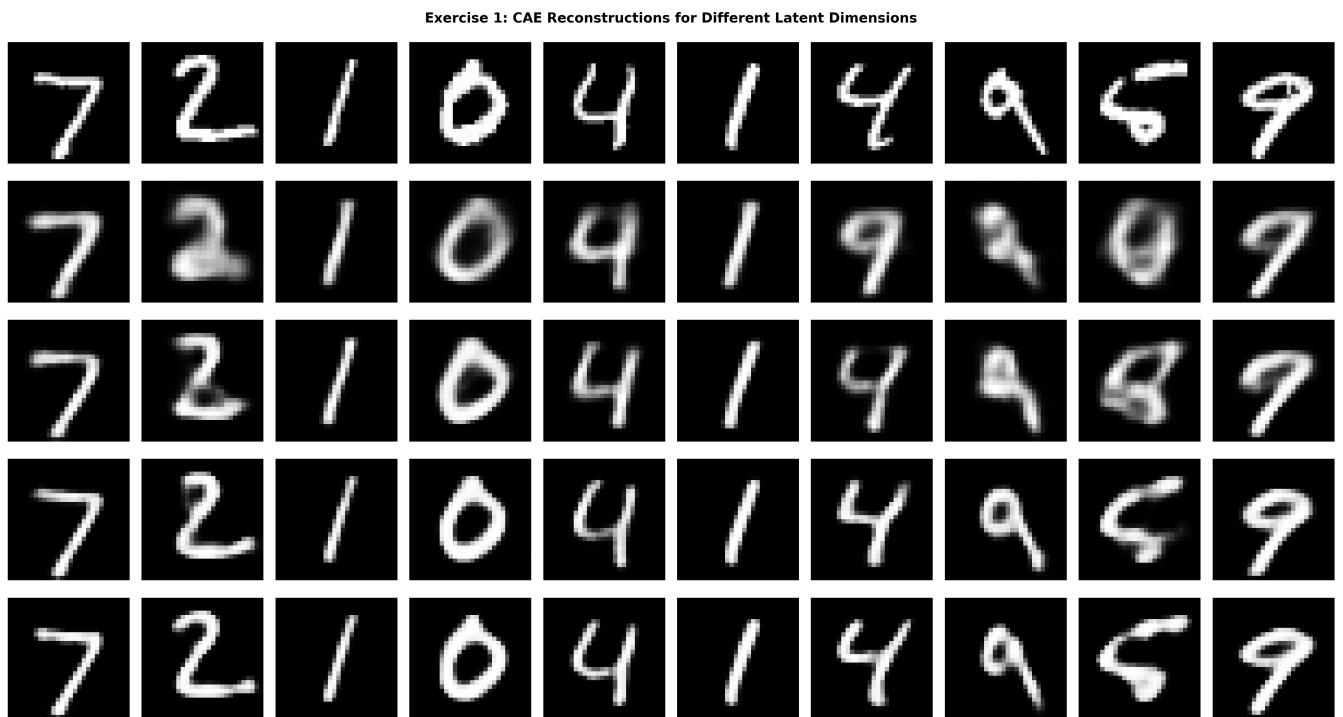


Figure 14: Reconstruction results of the convolutional autoencoder for different latent dimensions.

Inference: The reconstructions become clearer as the latent dimension increases, with the higher-dimensional models preserving more of the digit shapes and fine details. The difference is most noticeable at the smaller latent dimensions, where some strokes are more blurred or distorted. This shows that a larger latent space helps the CAE retain more information during reconstruction.

28.2 Exercise 2: Gaussian Noise vs. Salt-and-Pepper Noise

The same denoising convolutional autoencoder architecture was used to compare Gaussian noise and salt-and-pepper noise. Gaussian noise with $\sigma = 0.2$ and salt-and-pepper noise with $p = 0.10$ were considered.

Noise Type	MSE	MAE	SSIM
Gaussian ($\sigma = 0.2$)	0.00468	0.02138	0.9442
Salt-and-Pepper ($p = 0.10$)	0.00501	0.02152	0.94284

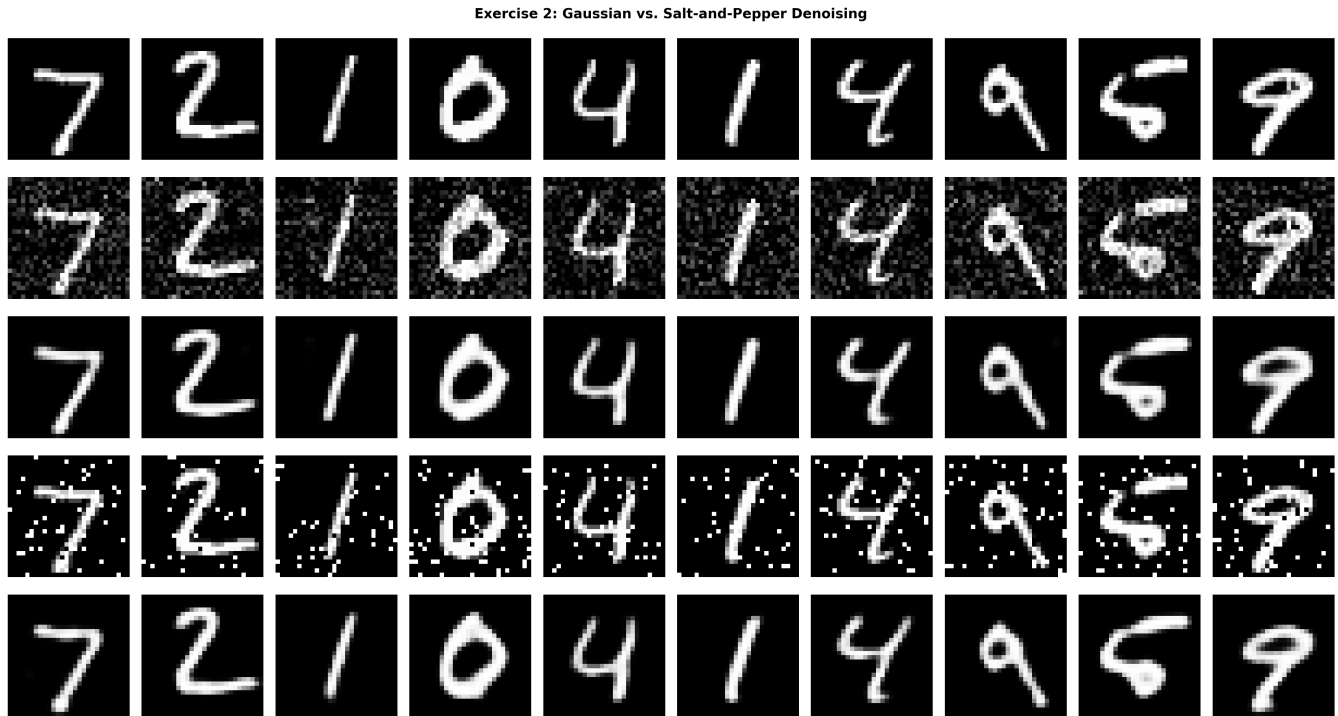


Figure 15: Comparison of denoising performance for Gaussian and salt-and-pepper noise.

Inference: The denoising autoencoder is able to recover the main digit structure from both Gaussian and salt-and-pepper noise. The denoised images are quite similar to the clean images in both cases, although some small noise and detail loss can still be seen. Overall, the model handles both types of corruption effectively.

28.3 Exercise 3: Effect of Noise Level on Denoising Quality

The denoising autoencoder was evaluated at two different Gaussian noise levels to study how the strength of corruption affects SSIM.

Gaussian Noise Level	SSIM
$\sigma = 0.1$	0.96068
$\sigma = 0.3$	0.91271

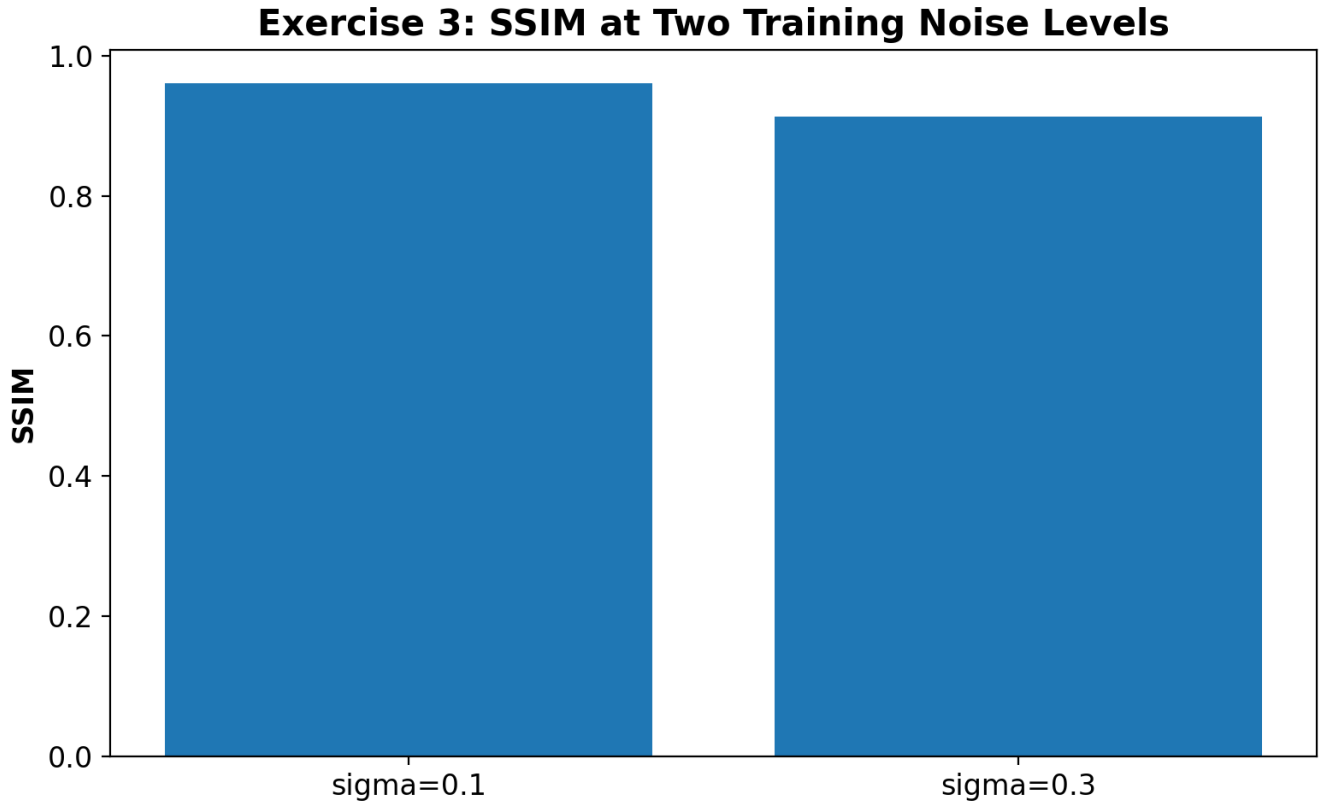


Figure 16: Comparison of SSIM at two Gaussian noise levels.

Inference:

The SSIM decreases as the noise level increases. It falls from 0.96068 at $\sigma = 0.1$ to 0.91271 at $\sigma = 0.3$, a decrease of 0.04797. This shows that stronger corruption makes it more difficult for the denoising autoencoder to recover the original image structure.

28.4 Exercise 4: Conv2DTranspose Decoder vs. UpSampling Decoder

The standard convolutional decoder using UpSampling2D was compared with a decoder using Conv2DTranspose.

Decoder	MSE	MAE	SSIM	Parameters
UpSampling2D	0.00259284	0.015106	0.973091	74497
Conv2DTranspose	0.00254852	0.014899	0.973346	74497

Exercise 4: CAE Decoder Comparison

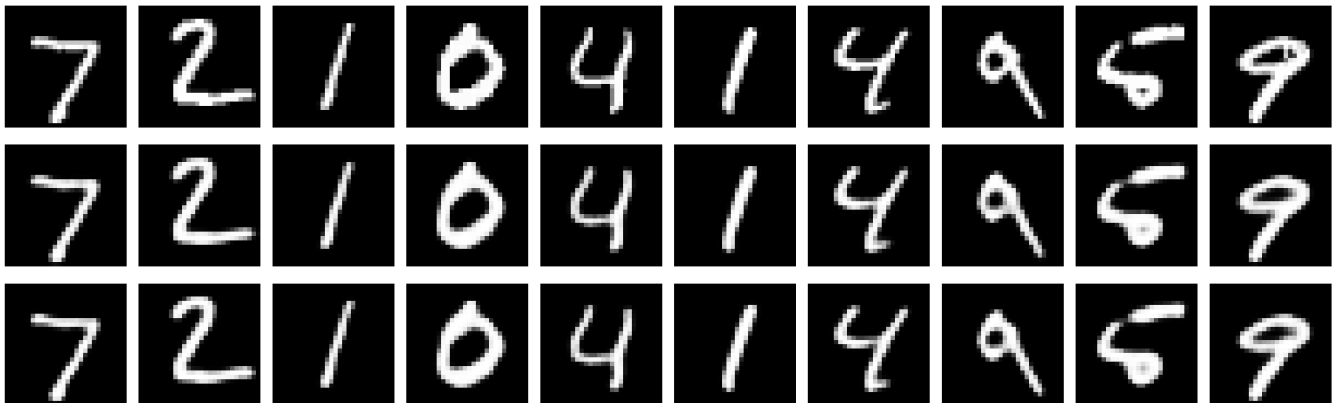


Figure 17: Comparison of CAE reconstructions using UpSampling2D and Conv2DTranspose decoders.

Inference: Both decoder architectures produce very similar reconstructions, with only small visual differences between the two. In this experiment, the Conv2DTranspose decoder performed slightly better because it had

slightly lower MSE and MAE and slightly higher SSIM than the UpSampling2D decoder. Both decoders have the same number of parameters (74497), and the overall reconstruction quality remains very close for both approaches.

28.5 Exercise 5: VAE with Latent Dimensions 2 and 8

Two VAE models with latent dimensions of 2 and 8 were compared to study the effect of latent-space size on reconstruction quality and representation.

Latent Dimension	Reconstruction Loss	KL Loss	MSE	SSIM
2	152.362	5.692	0.04492	0.5058
8	145.30267	7.90089	0.04053	0.53389

Exercise 5: VAE Latent Dimension 2 vs. 8

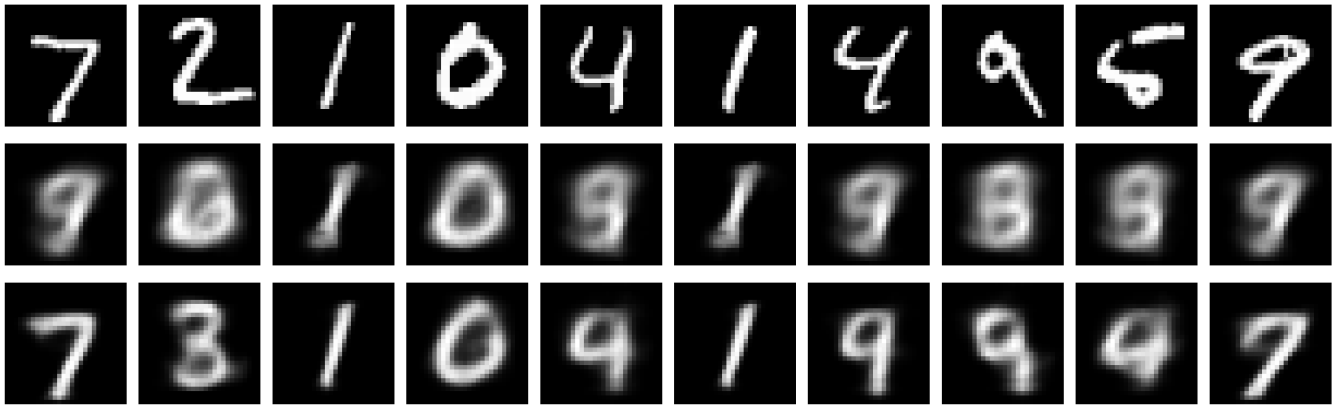


Figure 18: Comparison of VAE reconstruction performance for latent dimensions 2 and 8.

Inference: The VAE with latent dimension 8 produces clearer reconstructions than the 2-dimensional model, with more of the digit shapes and details being preserved. This is consistent with the lower MSE and higher SSIM, showing that the larger latent space can retain more information.

Exercise 5: VAE Latent Spaces

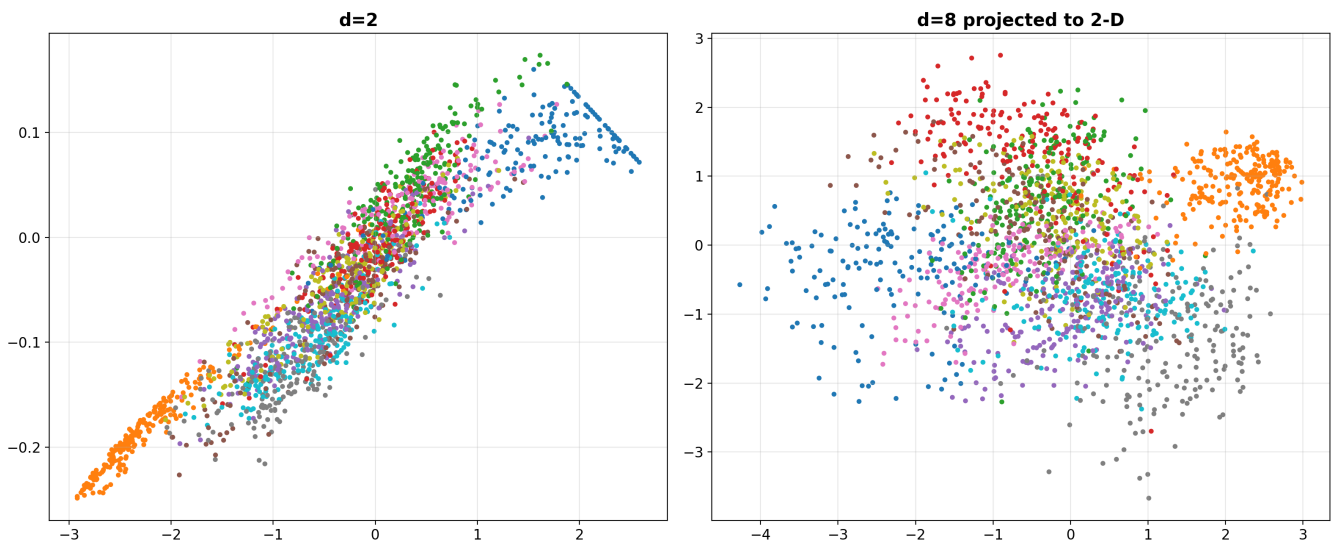


Figure 19: Latent-space representations for VAE models with different latent dimensions.

Inference: The 2-dimensional latent space shows a clear structured pattern with some overlap between digit classes. The 8-dimensional latent representation contains more room to capture variation between the digits, resulting in a more spread-out representation when projected to 2D. This shows that the larger latent space can capture richer representations of the input data.

28.6 Exercise 6: Effect of Increasing the KL-Loss Contribution

The effect of increasing the KL-loss contribution was studied by comparing a standard VAE with $\beta = 1$ and a model with a stronger KL weighting of $\beta = 5$.

Model	Reconstruction Loss	Weighted KL	MSE	SSIM
$\beta = 1$	177.36964	2.37935	0.05533	0.30906
$\beta = 5$	179.61571	7.67144	0.05566	0.29916

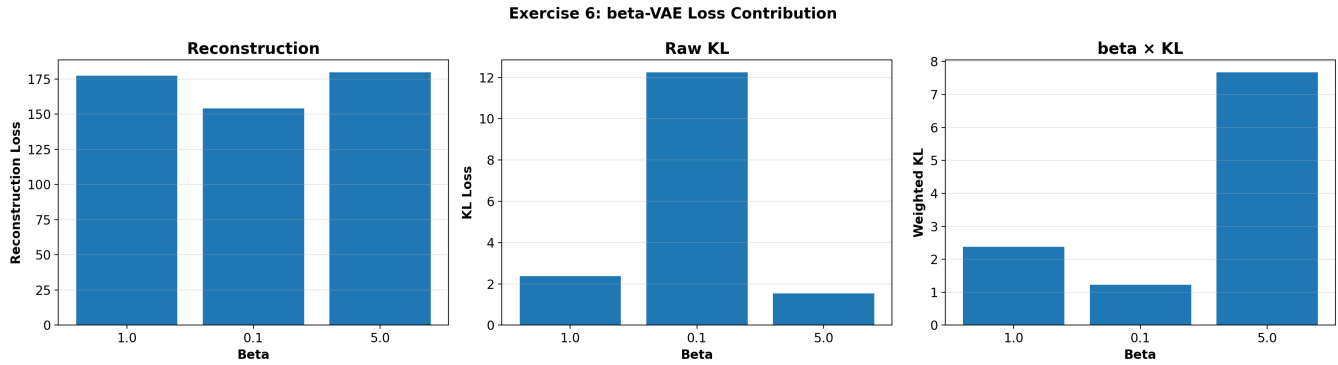


Figure 20: Loss comparison for VAE models with different KL-loss contributions.

Inference: Increasing β from 1 to 5 increases the KL regularization contribution (weighted KL rises from 2.37935 to 7.67144) and creates a trade-off with reconstruction quality. In this experiment, $\beta = 5$ has slightly higher reconstruction loss (179.61571 versus 177.36964) and slightly lower SSIM (0.29916 versus 0.30906) than $\beta = 1$. This shows that stronger KL regularization puts more constraint on the latent representation.

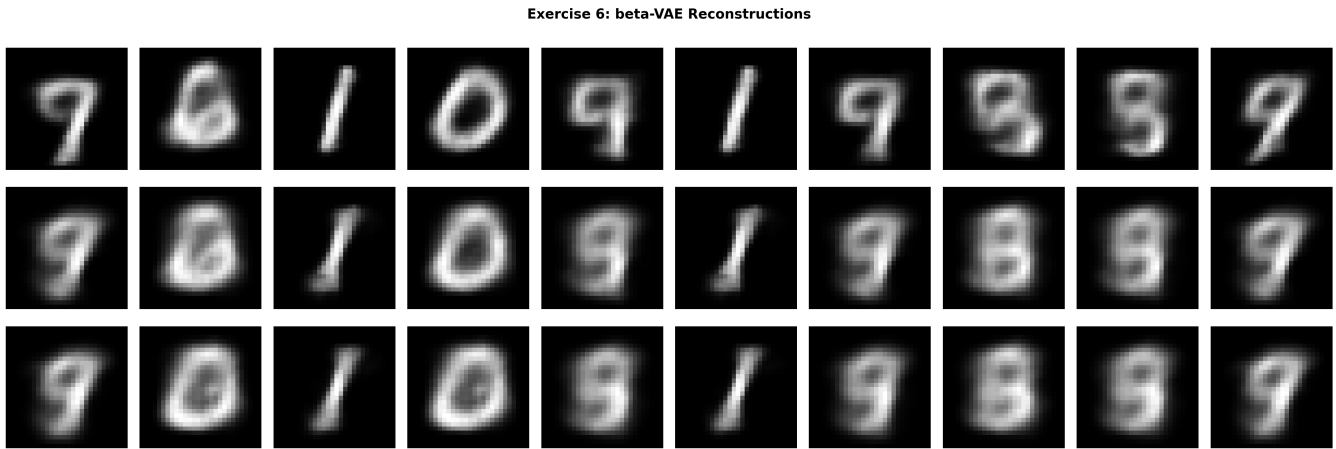


Figure 21: Reconstruction comparison for different KL-loss weights.

Inference: Both models are able to reconstruct the main shapes of the digits, but the $\beta = 5$ reconstructions appear slightly more blurred and lose some fine details compared with $\beta = 1$. This is consistent with the stronger KL regularization applied to the latent representation.

28.7 Exercise 7: Larger Grid of VAE Samples

A larger grid of generated images was produced from the VAE latent prior using a latent dimension of 8. A total of 64 samples were generated to observe the diversity of the outputs.

Exercise 7: 64 VAE Samples (d=8)

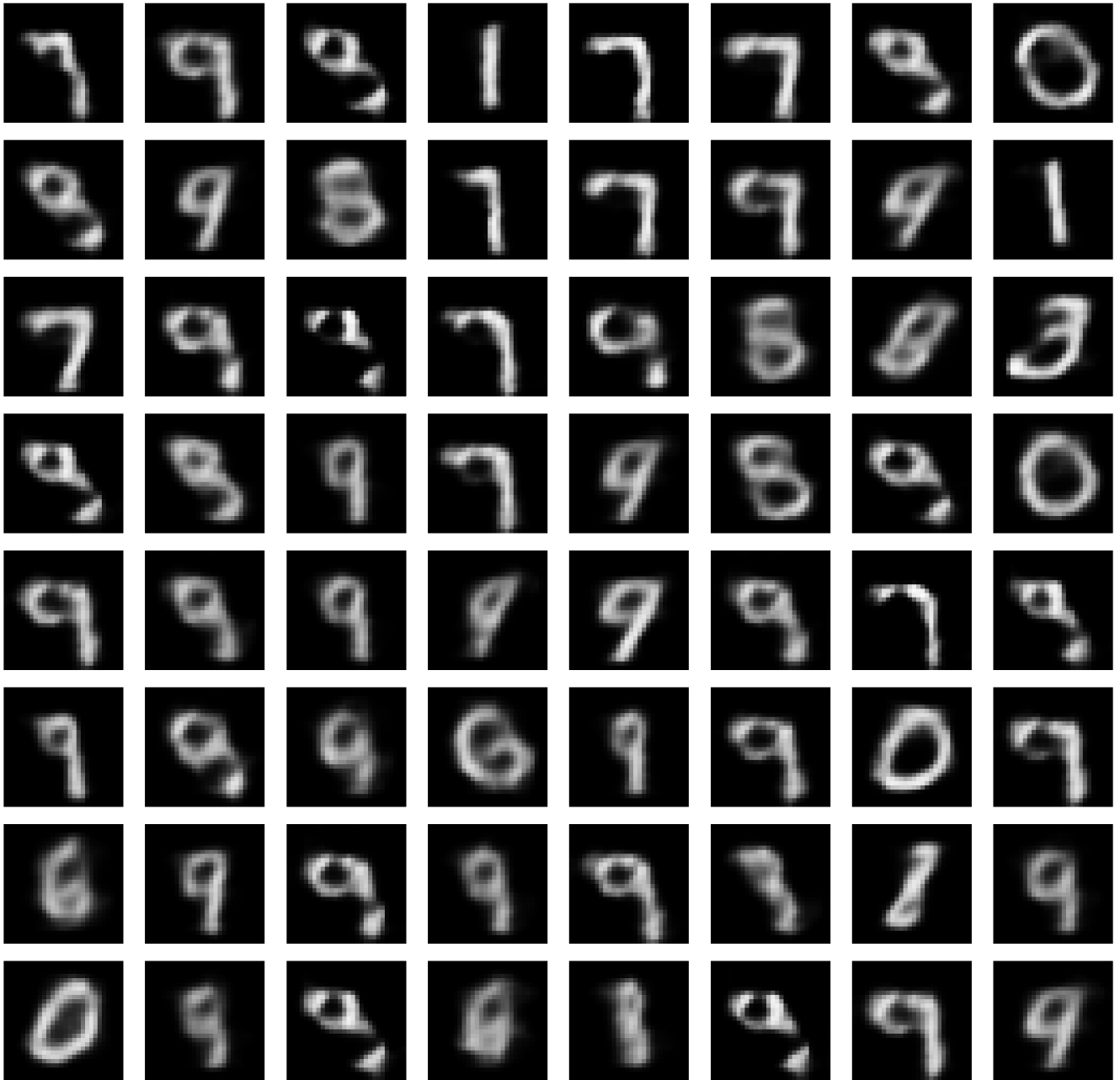


Figure 22: 64 VAE-generated MNIST samples using latent dimension 8.

Inference:

The generated samples show noticeable variation in digit shapes and stroke styles, indicating that the VAE can generate different images from randomly sampled latent vectors. Most samples retain recognizable digit-like structures, although some are still blurry or ambiguous. Overall, the larger grid shows that the VAE is able to generate a diverse set of new digit-like images.

28.8 Exercise 8: Latent-Space Interpolation Between Digit Regions

Two regions corresponding to different digit classes were selected in the two-dimensional VAE latent space. Latent vectors from the selected regions were then interpolated between the two endpoints.

$$z(\alpha) = (1 - \alpha)z_1 + \alpha z_2, \quad \alpha \in [0, 1].$$

Exercise 8: Digit 4 and 9 Latent Regions

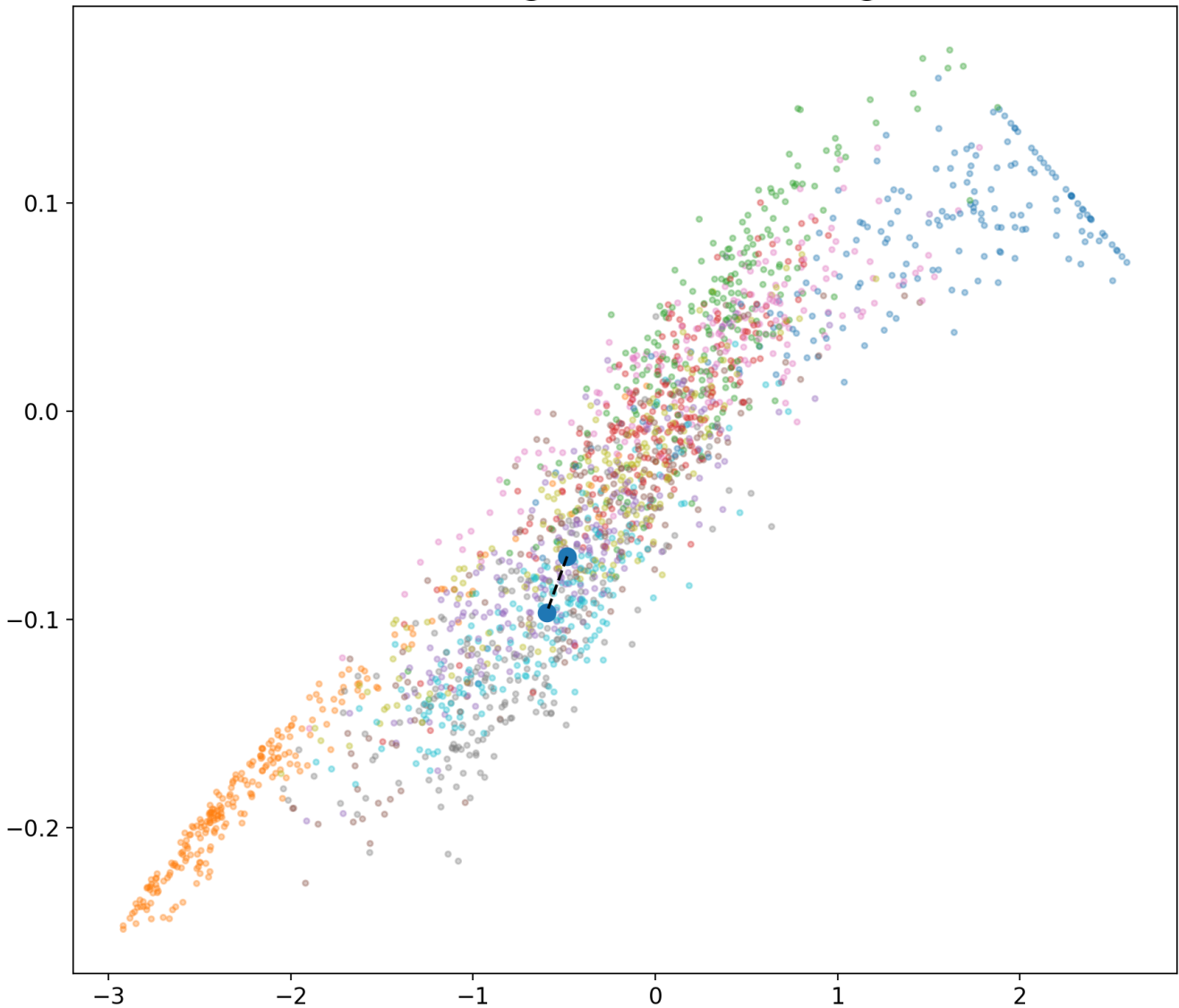


Figure 23: Selected digit regions in the two-dimensional VAE latent space.

Inference: The selected regions lie in different parts of the two-dimensional latent space and correspond to different digit patterns. This shows that the VAE latent space contains distinct regions that can be used as starting and ending points for interpolation.

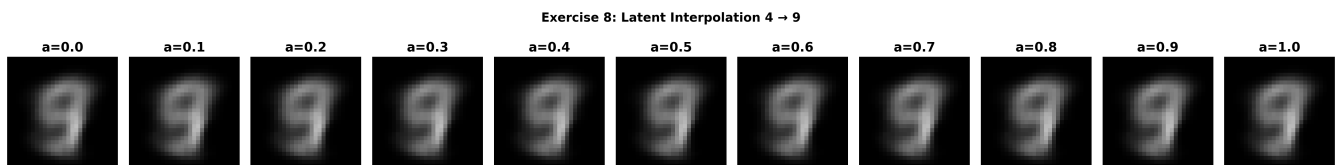


Figure 24: VAE interpolation between selected latent-space regions.

Inference: The generated images change gradually as the latent vector moves between the two selected regions. The intermediate outputs retain meaningful digit-like structures, showing that the VAE can produce smooth transitions between different latent representations.

29. Source Code

GitHub Repository

30. Expected Outcome

At the end of the experiment, students should be able to demonstrate the complete image-representation and reconstruction pipeline:

Image $\rightarrow$ Encoder $\rightarrow$ Latent Representation $\rightarrow$ Decoder $\rightarrow$ Reconstructed Image

They should also demonstrate denoising:

Clean Image $\rightarrow$ Noise $\rightarrow$ Denoising Autoencoder $\rightarrow$ Recovered Image

and variational generation:

Image $\rightarrow (\mu, \sigma) \rightarrow z \rightarrow$ Decoder $\rightarrow$ Generated/Reconstructed Image

The final report should contain both quantitative evaluation and visual analysis. Numerical performance values must be obtained from the student's own execution.

31. References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Diederik P. Kingma and Max Welling, "Auto-Encoding Variational Bayes," ICLR, 2014.
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5. TensorFlow Documentation: <https://www.tensorflow.org>
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7. scikit-image Documentation: <https://scikit-image.org>